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Which jobs will be in demand? Which ones are shrinking? And which ones could be hardest to fill?

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At a glance

- **During the pandemic (2019–22), the US labor market saw 8.6 million occupational shifts, 50 percent more than in the previous three-year period.** Most involved people leaving food services, in-person sales, and office support for different occupations.
- **By 2030, activities that account for up to 30 percent of hours currently worked across the US economy could be automated—a trend accelerated by generative AI.** However, we see generative AI enhancing the way STEM, creative, and business and legal professionals work, rather than eliminating a significant number of jobs outright. Automation's biggest effects are likely to hit other job categories. Office support, customer service, and food service employment could continue to decline.
- **Federal investment to address climate and infrastructure, as well as structural shifts, will also alter labor demand.** The net-zero transition will shift employment away from oil, gas, and automotive manufacturing and into green industries for a modest net gain in employment. Infrastructure projects will increase demand in construction, which is already short almost 400,000 workers today. We also see increased demand for healthcare workers as the population ages, plus gains in transportation services due to e-commerce.
- **An additional 12 million occupational transitions may be needed by 2030.** As people leave shrinking occupations, the economy could swing toward higher-wage jobs. Workers in lower-wage jobs are up to 14 times more likely to need to change occupations than those in highest-wage positions, and most will need additional skills to do so successfully. Women are 1.5 times more likely to need to move into new occupations than men.
- **The United States will need workforce development on a far larger scale as well as more expansive hiring approaches from employers.** Employers will need to hire for skills and competencies rather than credentials, recruit from overlooked populations (such as rural workers and people with disabilities), and deliver training that keeps pace with their evolving needs.

The US labor market is going through a rapid evolution in the way people work and the work people do. Months after MGI released its last report on the future of work in America, the world found itself battling a global pandemic.^[1] Since then, the US job market has come roaring back from its sudden drop. The nature of work has changed as many workers have shifted with remote or hybrid models and employers have sped up their adoption of automation technologies. More recently, the accelerated development of generative AI, with its advanced natural language capabilities, has extended the possibilities for automation to a much wider set of occupations.

Amid this disruption, workers changed jobs at a remarkable pace—and a subset made bigger leaps and moved into entirely different occupations (Exhibit 1). Some 8.6 million occupational shifts took place from 2019 through 2022. Now even more change is in store. We expect an additional 12 million occupational shifts by 2030. The total number of transitions through 2030 could be 25 percent higher than we projected a little over two years ago.^[2]

ENR1

Multiple forces are set to fuel growth in certain occupations and erode jobs in others. They generally fall into three categories: automation, including generative AI; an injection of federal investment into infrastructure and the net-zero transition; and long-term structural trends such as aging, continuing investment in technology, and the growth of e-commerce and remote work. We do not forecast how aggregated employment may be affected by the business cycle in the short term; instead, we focus on how these forces may reshape the composition of labor demand over the long term.

Across a majority of occupations employing 75 percent of the workforce, the pandemic accelerated trends that could persist through the end of the decade. Occupations that took a hit during the downturn are likely to continue shrinking over time. These include customer-facing roles affected by the shift to e-commerce and office support roles that could be eliminated either by automation or by fewer people coming into physical offices. Declines in food services, customer service and sales, office support, and production work could account for almost ten million (more than 84 percent) of the 12 million occupational shifts expected by 2030.

By contrast, occupations in business and legal professions, management, healthcare, transportation, and STEM were resilient during the pandemic and are poised for continued growth. These categories are expected to see fewer than one million occupational shifts by 2030.

“Workers have shown a willingness to change career paths, while a tighter labor market has encouraged companies to hire from broader applicant pools.”

For the other categories that account for the remaining one million occupational shifts still to come, the pandemic was a temporary headwind. Employment in fields like education and training should rise in the years ahead amid a continuous need for early education and lifelong learning. Demand for construction workers also stalled during the height of the pandemic but is expected to rebound strongly.

The changes estimated in our earlier research are happening even faster and on an even bigger scale than expected. It is becoming even more urgent to solve occupational and geographic mismatches and connect workers with the training they need to land jobs with better prospects. The fact that workers have been willing to pivot and change career paths, while a tighter labor market encouraged companies to hire from broader applicant pools, gives cause for optimism—but not complacency. The future of work is already here, and it's moving fast.

By the end of 2022, employment had bounced back to its 2019 level. But a great deal was in flux.

Are pandemic-era labor shortages here to stay?

The quits rate soared to new heights during the pandemic, with roughly 48 million Americans leaving their jobs in 2021 and 51 million in 2022. What people did next is not fully evident from the data. Some moved into better jobs with higher pay. Others left the labor force, whether out of discouragement or for personal or health reasons, and it is unclear if or when they will return.

Total employment hit an all-time high after the pandemic, with many employers encountering hiring difficulties. As of April 2023, some ten million positions remained vacant; labor force participation had ticked up but was 0.7 percentage point below its prepandemic level. That translates into roughly 1.9 million workers who are neither employed nor actively looking for jobs. This erosion comes after an extended 20-year trend of steadily falling participation.

Labor supply may continue to be constrained, given that one in four Americans will be of retirement age or older by 2030. Without higher participation rates, increased immigration, or meaningful productivity growth, labor shortages could be a lasting issue as the economy and the population grow. This remains an open question confronting markets, economists, and employers.

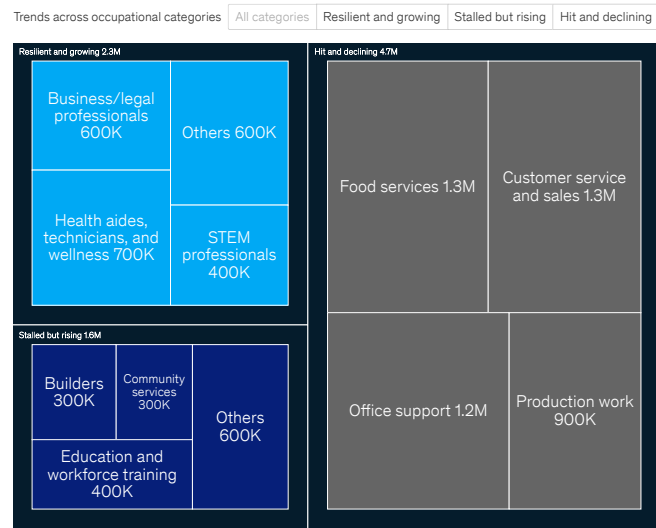
The Great Attrition obscured deeper shifts

While most attention was focused on soaring quits rates during the pandemic, something more structural was also occurring. A subset of people did more than change employers; they moved into different occupations altogether. Based on net increases and decreases in employment, some 8.6 million occupational shifts took place from 2019 through 2022—50 percent more than in the previous three-year period (Exhibit 2).^[3] While it is impossible to trace individual moves, many people left their previous roles and landed better-paying jobs in other occupations.

ENR2

More than half of the ~8.6 million recent occupational shifts in the United States involved workers leaving roles in food services, customer service, office support, and production.

Estimated shifts to another occupation, by category, 2019–22



Source: O*NET, US Bureau of Labor Statistics; Current Population Survey; US Census Bureau; McKinsey Global Institute analysis; McKinsey & Company

The majority of these shifts came from people leaving jobs in food services, customer service and sales, office support, and production work (such as manufacturing). At the same time, managerial and professional roles plus transportation services collectively added close to four million jobs from 2019 to 2022. Our previous research had anticipated these types of changes over a longer time frame, but the pandemic suddenly accelerated matters. The past few years have been a preview of trends we expect to continue through the end of the decade.

More high-wage jobs—and fewer workers taking lower-wage service jobs

Overall employment in low- and middle-wage occupations has fallen from prepandemic levels, while occupations that pay more than \$57,000 annually added about 3.5 million jobs. However, it is unclear how many higher-paying roles were filled by people who moved up and how many were filled by new entrants to the labor force. Meanwhile, the number of lower-wage job openings has not declined. Demand for lower-wage service work remains, but fewer workers are accepting these roles.

What is clear from the job switching and occupational shifts of the past three years is that the US labor market accommodated a higher level of dynamic movement. Spiking demand and labor scarcity forced many employers to consider nontraditional candidates with potential and train them if they lacked direct experience. While this may not hold in the future, employers and workers alike can draw on what they have learned about the potential for people to make quick pivots and add new skills.

Automation, from industrial robots to automated document processing systems, continues to be the biggest factor in changing the demand for various occupations. Generative AI is both accelerating automation and extending it to an entirely new set of occupations. While this technology is advancing rapidly, other forces are also affecting labor demand. Overall, we expect significant shifts in the occupational mix in the United States through the end of the decade.

The effects of automation and generative AI

Automation has taken a leap forward with the recent introduction of generative AI tools. “Generative” refers to the fact that these tools can identify patterns across enormous sets of data and generate new content—an ability that has often been considered uniquely

human. Their most striking advance is in natural language capabilities, which are required for a large number of work activities. While ChatGPT is focused on text, other AI systems from major platforms can generate images, video, and audio.

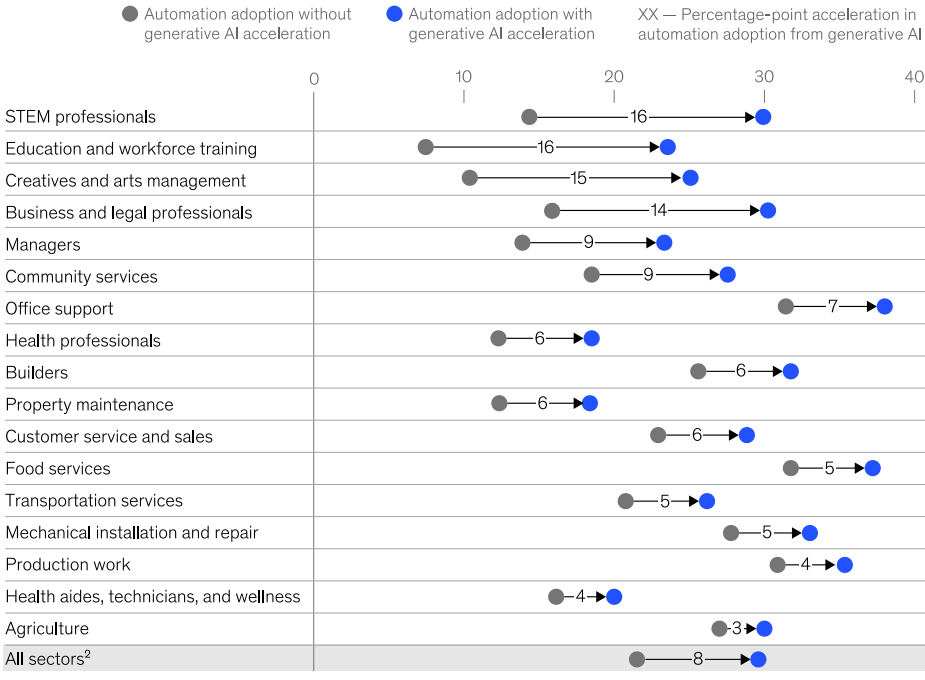
Although generative AI is still in the early stages, the potential applications for businesses are significant and wide-ranging. Generative AI can be used to write code, design products, create marketing content and strategies, streamline operations, analyze legal documents, provide customer service via chatbots, and even accelerate scientific discovery. It can be used on its own or with "humans in the loop," the latter is more likely at present, given its current level of maturity.

All of this means that automation is about to affect a wider set of work activities involving expertise, interaction with people, and creativity. The timeline for automation adoption could be sharply accelerated. Without generative AI, our research estimated, automation could take over tasks accounting for 21.5 percent of the hours worked in the US economy by 2030. With it, that share has now jumped to 29.5 percent (Exhibit 3).^[4]

Exhibit 3

With generative AI added to the picture, 30 percent of hours worked today could be automated by 2030.

Midpoint automation adoption¹ by 2030 as a share of time spent on work activities, US, %



¹Midpoint automation adoption is the average of early and late automation adoption scenarios as referenced in *The economic potential of generative AI: The next productivity frontier*, McKinsey & Company, June 2023.
²Tasks are weighted by 2022 employment in each occupation.
Source: D'NET, US Bureau of Labor Statistics, McKinsey Global Institute analysis

McKinsey & Company

Other forces affecting future labor demand

Automation is not occurring in a vacuum, of course. Other trends are affecting the demand for certain occupations, and we expect the employment mix to change significantly through 2030, with more healthcare, STEM, and managerial positions and fewer jobs in customer service, office support, and food services.

- **Federal investment:** Recent federal legislation is driving momentum and investment in other areas that will affect jobs.^[1] Reaching the net-zero emissions goal is one of these priorities. Some 3.5 million jobs could be displaced through direct and indirect effects across the economy. But at the macro level, these losses should be more than offset by gains of 4.2 million jobs, primarily led by capital expenditures on renewable energy. The net-zero transition will likely be a net positive for jobs, but those jobs may be located in different places and require different skills.

Similarly, major investment in infrastructure projects across the country will bolster construction jobs, which could see employment growth of 12 percent from 2022 through 2030. However, the sector already had some 383,000 unfilled positions in April 2023. This shortage will have to be addressed to bring infrastructure projects to life from coast to coast.^[2]

The CHIPS and Science Act is putting additional funding into semiconductor manufacturing as well as R&D and scientific research.^[3] This comes at a time when some companies have been adjusting their supply chains, leading to an uptick in domestic manufacturing. While manufacturing is likely to boost employment demand overall in the years ahead, the sector is becoming more high-tech. It will involve fewer traditional production jobs than in the past but more workers with technical and STEM skills.^[4]

- **Other structural trends:** At the same time, other trends like rising incomes and education levels will sustain jobs. An aging population will need more healthcare workers in multiple roles, while the ongoing process of digitizing the economy will require adding more tech workers in every sector.

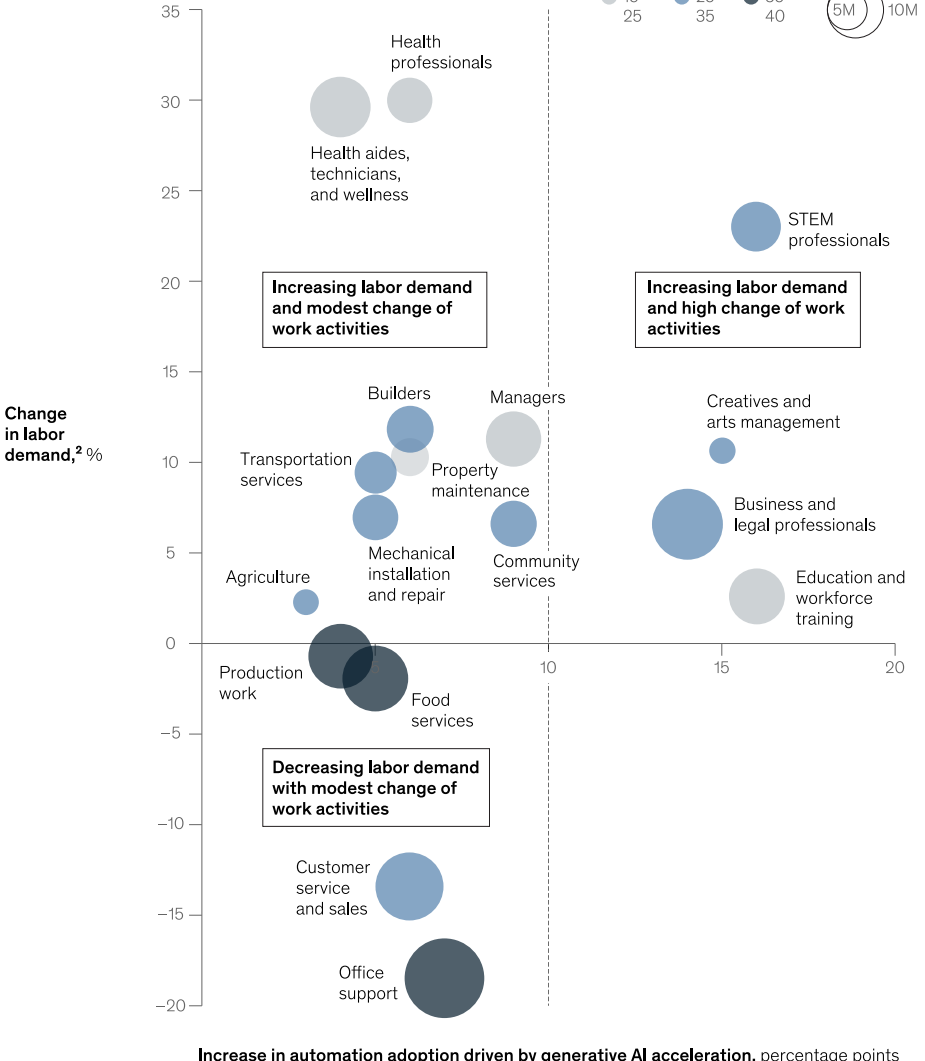
One of the biggest questions of recent months is whether generative AI might wipe out jobs. Our research does not lead us to that conclusion, although we cannot definitively rule out job losses, at least in the short term. Technological advances often cause disruption, but historically, they eventually fuel economic and employment growth.

This research does not predict aggregated future employment levels; instead, we model various drivers of labor demand to look at how the mix of jobs might change—and those results yield some gains and some losses.^[1] In fact, the occupational categories most exposed to generative AI could continue to add jobs through 2030 (Exhibit 4), although its adoption may slow their rate of growth. And even as automation takes hold, investment and structural drivers will support employment. The biggest impact for knowledge workers that we can state with certainty is that generative AI is likely to significantly change their mix of work activities.

Exhibit 4

While STEM, healthcare, builders, and professional fields continue to add jobs, generative AI could change work activities significantly for many occupations.

Estimated labor demand change and generative AI automation acceleration by occupation, US, 2022–30



¹Midpoint automation adoption is the average of early and late automation adoption scenarios as referenced in *The economic potential of generative AI: The next productivity frontier*, McKinsey & Company, June 2023.
We consider multiple drivers affecting demand: rising income, aging populations, technology investment, infrastructure investment (including Bipartisan Infrastructure Law), rising education levels, net-zero transitions, monetization of unpaid work, creation of new occupations, automation (including generative AI), increased remote working and virtual meetings, and e-commerce and other virtual transactions.
Source: US Bureau of Labor Statistics, Current Population Survey, US Census Bureau, McKinsey Global Institute analysis

McKinsey & Company

Resilient and growing occupational categories

The largest future job gains are expected to be in healthcare, an industry that already has an imbalance, with 1.9 million unfilled openings as of April 2023. We estimate that there could be demand for 3.5 million more jobs for health aides, health technicians, and wellness workers, plus an additional two million healthcare professionals.^[1]

By 2030, we further estimate a 23 percent increase in the demand for STEM jobs. Although layoffs in the tech sector have been making headlines in 2023, this does not change the longer-term demand for tech talent among companies of all sizes and sectors as the economy continues to digitize. Employers in banking, insurance, pharmaceuticals, and healthcare, for example, are undertaking major digital transformations and need tech workers with advanced skills.^[2] In addition, the transportation services category is expected to see job growth of 9 percent by 2030.

Declining occupational categories

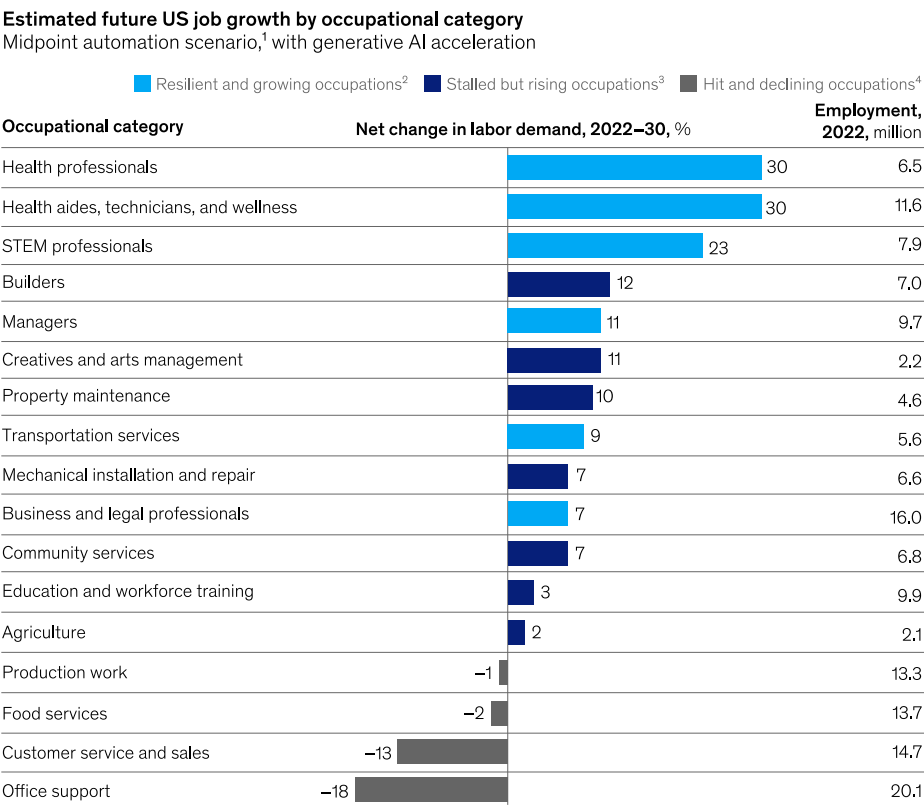
The biggest future job losses are likely to occur in office support, customer service, and food services. We estimate that demand for clerical^[3] could decrease by 1.8 million jobs, in addition to losses of 830,000 for retail salespersons, 710,000 for administrative assistants, and 630,000 for cashiers. These jobs involve a high share of repetitive tasks, data collection, and elementary data processing, all activities that automated systems can handle efficiently. Our analysis also finds a modest decline in production jobs despite an upswing in the overall US manufacturing sector, which is explained by the fact that the sector increasingly requires fewer traditional production jobs but more skilled technical and digital roles.^[4]

We estimate that 11.8 million workers currently in occupations with shrinking demand may need to move into different lines of work by 2030. Roughly nine million of them may wind up moving into different occupational categories altogether. Considering what has already transpired, that would bring the total number of occupational transitions through the decade's end to a level almost 25 percent higher than our earlier estimates, creating a more pronounced shift in the mix of jobs across the economy.

Overall, we expect more growth in demand for jobs requiring higher levels of education and skills, plus declines in roles that typically do not require college degrees (Exhibit 5).

Exhibit 5

Healthcare, STEM, and builder roles could grow, while demand for office support and customer service roles could decline.



¹Support automation adoption is the average of early and late automation adoption scenarios as referenced in *The economic potential of generative AI: The next productivity frontier*, McKinsey & Company, June 2023.
²Resilient during the pandemic, 2019–22, and expected to grow between 2022 and 2030.
³Stalled during the pandemic, 2019–22, and expected to rise between 2022 and 2030.
⁴Hit during the pandemic, 2019–22, and continuing to decline between 2022 and 2030.
Source: CPBETI, US Bureau of Labor Statistics, Current Population Survey, US Census Bureau, McKinsey Global Institute analysis

McKinsey & Company

People in the two lowest wage quintiles (those earning less than \$30,800 a year and those earning \$30,800 to \$36,200 a year) are up to 10 and 14 times more likely, respectively, to need to change occupations by the end of this decade than the highest earners.¹⁴ Changing occupations, as opposed to finding a new job within the same occupation, often requires adding new skills and is more challenging.

The jobs in the two lowest wage quintiles are disproportionately held today by those with less education, women, and people of color. Women are heavily represented in office support and customer service, which could shrink by about 3.7 million and 2.0 million jobs, respectively, by 2030. Similarly, black and hispanic workers are highly concentrated in some shrinking occupations within customer service, food services, and production work.

While our analysis shows a decrease of 11 million jobs in the two lowest wage quintiles by 2030, jobs in the highest wage quintile could grow sharply, by 3.8 million. Helping workers in lower-wage, shrinking occupations move into better-paying jobs with more stability will require widespread access to training programs, effective job matching, different hiring and training practices by employers, and better geographic mobility.

The overall labor market will have higher demand for social-emotional and digital skills. Although the demand for basic cognitive and manual skills is likely to decline, physical work is not going away. It may still account for just under 31 percent of time spent, driven by growth in sectors such as transportation services, construction, and healthcare.

With the pace of change unlikely to let up, the challenge will be helping workers match up with the jobs of the future. While some of this may require large-scale collaboration, individual companies can fill many of the gaps by adapting their own approaches to hiring and training.

Boosting productivity through automation and generative AI

Recent MGI research focused on how to reignite productivity growth in the United States.¹⁵ Automation and reskilling will be vital to this effort. Automation could jump-start lackluster productivity while simultaneously easing labor shortages.

Generative AI has the potential to increase US labor productivity by 0.5 to 0.9 percentage points annually through 2030 in a midpoint adoption scenario. The range reflects whether the time freed up by automation is redeployed at 2022 productivity levels or 2030 levels, with both scenarios accounting for the occupational mix expected in 2030.

Combining generative AI with all other automation technologies, the potential growth could be even larger. All types of automation could help drive US productivity growth to 3 to 4 percent annually in a midpoint adoption scenario. However, this will require significant action from stakeholders across the public and private sector. Workers will need support in learning new skills, and other risks associated with generative AI also need to be mitigated and controlled. But if worker transitions and risks are well managed, generative AI could contribute substantively to economic growth.

To capture the full benefits of generative AI to make knowledge work more productive, employers, policy makers, and broader ecosystems would need to establish clear guidelines and guardrails—and workers would need to see these tools not as job destroyers but as work enhancers. When machines take over dull or unpleasant tasks, people can be left with more interesting work that requires creativity, problem-solving, and collaborating with others. Workers will need to gain proficiency with these tools and, importantly, use the time that is freed up to focus on higher-value activities. When managers automate more of their administrative and reporting tasks, for example, they can spend more time on strategic thinking and coaching. Similarly, researchers could speed up projects by relying on automation tools to sort and synthesize large data sets.

For employers, doubling down on innovative hiring strategies

Most employers can benefit from using a broader lens in hiring. Instead of insisting on prior experience that matches the responsibilities of an open role as closely as possible, organizations can evaluate candidates on their capacity to learn, their intrinsic capabilities, and their transferable skills.

“Hiring people with potential and training them on the job can be an answer to labor shortages—and give opportunities to people who need them.”

A great deal of skills development happens on the job. Previous MGI research found that work experience contributes 40 percent of the average individual's lifetime earnings in the United States.¹⁶ Skills learned through work experience are an even bigger determinant for people without educational credentials who start out in lower-wage work.

Filling the jobs of the future is an opportunity to make the labor market more inclusive. Employers may need to reconsider whether some credential requirements are really necessary. Some 60 percent of US workers have skills gained through experience but lack four-year college degrees. Initiatives like Tear the Paper Ceiling are supporting workers who have experience but not degrees by raising awareness among employers and providing resources.

Employers can also recruit from populations that are often overlooked, such as retirees who want to return to work, people with employment gaps, and the formerly incarcerated. Remote work, for example, is opening up long-needed opportunities for people with disabilities who cannot commute and those in rural communities.

Tackling other structural issues

Women left the workforce in relatively higher numbers than men during the pandemic. It took three full years for the number of working women in the United States to fully bounce back. Many women doing lower-wage work have family obligations that may leave them feeling that they can't take the risk of going back to school or trying a new occupation. Beyond the hiring practices that can encourage and enable women to make career transitions, the need for affordable childcare remains a major barrier.¹⁷ To address it, a number of private-sector employers are expanding childcare benefits, while some state and local governments are providing tax credits, subsidies, or direct funding. In addition, historically male-dominated fields such as construction that are facing labor shortages can fill those gaps with more women, improving diversity in the process.

One key area of job demand is in caregiving, which is critical social infrastructure. We anticipate that the two fastest-growing occupations through the end of this decade will be nurses and home healthcare aides.¹⁸ Childcare workers, as noted above, provide a vital service to working families. But people have been leaving these types of jobs in droves. Meeting these growing needs will likely hinge on upgrading the quality of what are today typically low-paying jobs with little security or advancement opportunities.

While large employers may be able to handle their own training needs, the magnitude of the reskilling challenge for the nation as a whole calls for broader partnerships with industry groups, educational providers, and nonprofits as well as incentives for investing in human capital. Addressing the need for reskilling with efforts beyond individual companies would help spread the cost, addressing the concerns of employers who might be reluctant to invest in training for employees who can subsequently leave.

With millions of jobs potentially being eliminated by automation—and even more being created in fields requiring different skills—the United States needs broad access to effective training programs as well as job-matching assistance that can help individuals find opportunities. Many initiatives are in place, but it will be critical to dramatically scale up what works and take a proactive approach to filling key shortages. One promising solution, still in the early stages, involves digital learning and employment records—a kind of digital microcredential that can document how an individual worker has acquired skills and also translate across companies and over time.

The US labor market has been remarkably resilient in the face of recent challenges and rapid changes. That kind of adaptability is exactly what it will take to navigate the next chapter as well, supporting individuals while helping businesses meet their talent needs so they can continue driving growth.

How relevant and useful is this article for you?

☆☆☆☆

1. *The future of work in America: People and places, today and tomorrow*, McKinsey Global Institute, July 2019.
2. *The future of work after COVID-19*, McKinsey Global Institute, February 2021.
3. Measured as net job losses for individual occupations across sectors, net of estimated retirements, derived from US Bureau of Labor Statistics (BLS) data. An administrative assistant who takes a similar position with another employer has simply switched jobs and is not part of this analysis. If that person becomes an office manager, they have changed occupations within the same category (office support). If they become a computer systems analyst, they have moved into a different occupational category (IT/STEM professionals). The latter two moves are the kind of occupational shifts that we measure. Since we are unable to trace exactly how individual workers move, we use net declines as a broad proxy in our forward-looking scenario; we refer to people needing to make transitions if demand is projected to decline in their current occupation.
4. Note that this is the midpoint, representing the average of a very wide range, from 2.5 to 35.3 percent.
5. While our scenario includes the impact of federal investment in the net-zero transition and infrastructure, it does not include the full impact of the CHIPS and Science Act and the Inflation Reduction Act, since representation extended outside of the time of this analysis. However, both pieces of legislation point to the possibility of additional upside.
6. Sam Hurnsman, Ash Kumar, and Ryan Lally, “AI’s labor growth demand plans to upgrade US infrastructure,” *McKinsey & Company*, October 2022.
7. Note that both the CHIPS and Science Act and the Inflation Reduction Act create room for additional upside in employment. But since there is still uncertainty about their implementation as of this writing, their effects on jobs are not explicitly incorporated into our scenario.
8. For more on this topic, see Richard Hahn, Gaurav Bhat, and Nita Sarin-Bhatia, “The future economy: New industrial technology can create a better, faster, stronger America,” Public Affairs, 2022.
9. We rely on employment projections from the US Bureau of Labor Statistics for 2020 employment levels.
10. Note that registered nurses, nurse practitioners, and nurse anesthetists are in the healthcare professionals category; nurse midwives and licensed practical and vocational nurses are in the health aides category.
11. Jan Siewert, “As Big Tech rubs victory, other industries are desperate to hire them,” *MarketWatch*, February 16, 2023, and Steve Luhn and Tappi Moake, “As Silicon Valley renews a tech talent shift accelerates,” *New York Times*, December 26, 2022.
12. Note that clerks include mailroom and information clerks, general office clerks, bookkeeping, accounting, and auditing clerks, and shipping, receiving, and inventory clerks.
13. *Building a more competitive US manufacturing sector*, McKinsey Global Institute, April 2021.
14. Workers in the second-lowest wage quintile (earning \$30,800 to \$36,200 per year) have an even higher risk of needing to transition occupations than the lowest-wage workers, who earn less than \$30,800 a year. Placing the lowest-wage workers with technology may not make economic sense, but at a certain wage level, the equation changes. In addition, some lower-wage jobs involve unpredictable physical work or customer-facing work that does not lend itself well to automation.
15. *Reinvigorating US productivity for a new era*, McKinsey Global Institute, February 2023.
16. *Human capital of work: The value of experience*, McKinsey Global Institute, June 2022.
17. Sarah Gelin, Ayushi Durrnell, Anika Kirovovich, and Kunal Modi, “The childcare conundrum: How can companies ease working parents’ return to the office?,” McKinsey & Company, May 2022.
18. For occupations that employed more than 50,000 people as of 2022.

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